

Name	Year	Purpose of the Study	Methodology	Limitations	Relevance to Our Work
Adaptive Traffic Signaling Control Using SUMO Simulator	International Journal of Innovative Science and Research Technology 2025	To develop a two-stage, three-module fuzzy logic system for real-time management of signalized junctions to reduce average vehicle waiting times.	Implemented a fuzzy logic traffic management system using the SUMO simulation tool with real-world traffic data from a congested intersection in Kilis, Turkey. The system dynamically adjusts signal timing using 52 fuzzy rules and was compared against conventional fixed-time control.	The study concentrates exclusively on an isolated intersection rather than a multi-intersection network.	Proves dynamic signal control in SUMO outperforms fixed-time systems. Uses real-time vehicle counts and queue lengths to adjust timings.
A SUMO Based Simulation Framework for Intelligent Traffic Management System	Journal of Traffic and Logistics Engineering Vol. 8, No. 1 2020	To explore a Deep-Neuro-Fuzzy model within the SUMO traffic simulation environment and develop a GUI to control simulation input attributes for optimal route planning.	Utilized C-means clustering and a Deep-Neuro-Fuzzy framework to generate dynamic road weights based on environmental and artificial attributes. Applied routing algorithms (Dijkstra, A*, CHWrapper) in SUMO to find optimal vehicle paths.	The model has not yet been implemented in real-time physical road networks, which would require costly personalized infrastructure. Other hidden parameters of existing SUMO algorithms remained unchanged during testing.	Shows how to integrate SUMO with an external backend to update traffic logic. Validates extracting live metrics and sending custom commands back to SUMO
AI-Powered Traffic Signal Control: A Deep Reinforcement Learning Framework for Urban Congestion Reduction	International Journal of Environmental Sciences 2025	To explore AI-based traffic signal regulation using deep reinforcement learning to minimize vehicle waiting times, maximize flow efficiency, and enhance pedestrian and emergency vehicle safety..	Formulated traffic signal optimization as a sequential decision-making task using DRL techniques like Proximal Policy Optimization (PPO), Deep Q-Networks (DQN), and Advantage Actor-Critic (A3C). Performed simulations using SUMO with real-world and synthetic data to compare AI methods against fixed-timing controls.	Relied on simulation-based testing, which did not fully account for erratic driving habits, infrastructural limitations, or external weather factors	Highlights using SUMO to prioritize emergency vehicles and specific vehicle classes. Confirms the method of using live queue data to optimize traffic lights.

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Enhancing Traffic Monitoring Systems through Image Analysis and Rule-Based Machine Learning	International Conference on Machine Learning and Data Engineering 2025	To present an advanced traffic monitoring system integrating reasoning-based image analysis and rule-based machine learning to improve traffic management, reduce congestion, and enhance road safety.	Leverages visual data from spot sensors utilizing a three-tier module approach. It processes daytime scenes via Google cloud using the RCNN algorithm and nighttime scenes via bimodal histograms and Otsu's thresholding for headlight extraction. A high-level module applies rule-based reasoning (via an inference engine and knowledge base) to refine the models.	Real-world testing was restricted to favorable weather, meaning the system's robustness in environmental conditions like fog or snow remains untested.	It proves that adding rule-based reasoning helps systems instantly adapt to unpredictable, real-time traffic scenarios.
Development of the Rule Based Approach to Traffic Management	2017 (the paper references the 2017 methodology and evaluation work)	To develop a Rule-Based Traffic Management (RBTM) approach that can manage traffic dynamically, detect congestion, handle conflicts, and distribute traffic across the road network more efficiently.	The study uses business rules, decision tables, traffic data, road-priority rules, and multi-agent concepts. The system detects traffic conditions such as saturation, congestion and gridlock, then requests actions such as increasing outbound flow, limiting inbound flow, or rerouting traffic.	The approach can shift traffic problems to lower-priority roads, increasing waiting queues there. It also caused an increase in red-light runners, which was an undesirable result. The second pilot did not use rerouting because suitable signs were unavailable.	This research helps in detecting traffic congestion and managing traffic automatically. It provides ideas for rule-based decision-making and traffic control. These concepts can help us improve traffic flow and reduce congestion.
Traffic Intersection Simulation Using Turning Movement Count Data in SUMO: A Case Study of Toronto Intersections	2025 — the paper was posted on arXiv on 14 August 2025	To develop CrossFlow , a tool that uses real-world traffic data to create realistic intersection simulations in SUMO , reducing the need for synthetic or randomly generated traffic data.	Real-world Turning Movement Count (TMC) data from Toronto was converted into SUMO-compatible traffic flows. A Python-based CrossFlow GUI was used to create the network, generate vehicle flows, and run simulations.	The study mainly focuses on regular 3-way and 4-way intersections. Pedestrian and bicycle data were not included, and individual vehicle-level control is limited because the study uses aggregated flow data.	It helps in real-time traffic analysis and congestion detection using real traffic data. It also provides a simulation-based approach for testing traffic signal optimization and improving traffic flow.

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Rule-Based Dynamic Traffic Management for Emergency Vehicle Routing: A Smart Infrastructure Approach	2019 — Published in June 2019	To develop an intelligent traffic management system that gives priority to emergency vehicles , dynamically adjusts traffic signals, and optimizes routes using real-time traffic, IoV, weather, and road-network data.	The proposed Rule-Based Dynamic Traffic Management (RBD-TM) combines real-time data collection, IoV/IoT sensors, GPS, RFID, shortest-path algorithms such as Dijkstra/Floyd-Warshall , GNNs, and dynamic traffic-signal control.	The paper does not clearly provide a separate limitations section. However, the proposed system depends on real-time sensors, IoV communication, GPS/RFID, and traffic infrastructure , which are required for effective operation.	This research helps in detecting traffic congestion and managing traffic dynamically . It provides ideas for emergency vehicle priority, automatic signal control, and route optimization , which can help improve traffic flow and reduce response time.
Dynamic Emergency Vehicle Prioritization in a Connected Urban Environment: A Rule-Based Approach	Published in June 2024	To develop an Emergency Vehicle Priority Management (EVPM) system that dynamically adjusts traffic signals and vehicle lanes to give priority to emergency vehicles while minimizing disruption to normal traffic.	The system uses a rule-based algorithm with IoV, 5G, GPS, V2X, RFID and sensors. It detects and tracks emergency vehicles, clears their lane, and changes traffic signals to green when required. The system was tested using SUMO with real-world TAPASCologne traffic data.	The study identifies high initial costs and data reliability as challenges. It also notes that lane clearance may not work optimally during severe congestion.	This research helps in dynamic traffic management and emergency vehicle prioritization . It provides useful ideas for automatic signal control, lane clearance, real-time traffic monitoring, and rule-based decision-making .
Traffic Intersection Simulation Using Turning Movement Count Data in SUMO: A Case Study of Toronto Intersections	2025 — posted on arXiv on 14 August 2025.	To develop CrossFlow , a tool that uses real-world Turning Movement Count (TMC) data to create realistic traffic simulations in SUMO , instead of using synthetic or randomly generated data.	Real-world traffic data from Toronto was converted into SUMO-compatible vehicle flows . A Python-based CrossFlow GUI was used to create the road network, generate traffic flows, and run simulations.	The study mainly focuses on 3-way and 4-way intersections . Pedestrian and bicycle data were outside the study scope, and individual vehicle-level control is limited because aggregated flow data is used.	This research helps in real-time traffic analysis and congestion management . It provides useful ideas for traffic simulation, signal optimization, and improving traffic flow using real-world data.

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AI-based traffic control and management systems – a synthesis of Macedonian research experiences	2025 — published in <i>Transportation Research Procedia</i>	The study aims to explore the use of AI and Machine Learning in traffic management. It focuses on reducing traffic congestion and vehicle emissions while improving traffic flow and throughput.	The study uses two case studies . The first applies Reinforcement Learning with Q-learning for adaptive traffic signal control using VISSIM simulation, while the second uses AI-based video processing to collect and analyze traffic data.	The study mainly relies on simulation-based testing , so real-world implementation requires further validation. Some traffic video data was also incomplete because of unsuitable camera positioning.	This research is relevant because it provides useful approaches for AI-based traffic monitoring and intelligent traffic control . The use of Q-learning, video analysis, and real-time traffic data can provide a foundation for developing our proposed traffic-management system.
Computation and Optimization of Traffic Network Topologies Using Eclipse SUMO.	The research paper was published in 2022 in <i>Annals of Emerging Technologies in Computing (AETiC)</i>	The study aims to analyze and optimize urban traffic network topology using traffic simulation. It specifically compares one-way and two-way street configurations to understand their impact on traffic congestion, travel time, traffic flow, and road capacity.	The researchers used Eclipse SUMO , a microscopic traffic simulation tool, to model traffic in Downtown Brickfields, Kuala Lumpur. They simulated one-way and two-way street networks and evaluated traffic speed, flow rate, density, and trip completion time under different traffic conditions.	The study is mainly based on computer simulation rather than real-time implementation. The results are specific to the selected road network and traffic scenarios, so different locations and traffic conditions may produce different results. The study also focuses mainly on street topology rather than complete intelligent traffic-control systems.	This research is relevant because it shows how traffic simulation and network optimization can be used to understand and reduce congestion. Eclipse SUMO, traffic-flow analysis, and comparison of different road configurations can provide a useful foundation for our proposed intelligent traffic-management system.